

Supplementary Table 1. Data Dictionary		
Biochemical Parameter	Label	Values
Data	Survey date	None
NO	ID number	None
VAR00001	None	None
Gender	Gender	0 = Female, 1 = Male
Age	Age	None
Age3Group	Age group (3 categories)	1 = '< 40', 2 = '≥40 and <60', 3 = '≥60'
AgeGroup	Age group	1 = '18-29y', 2 = '30-39y', 3 = '40-49y', 4 = '50-59y', 5 = '≥60y'
Height	Height	None
Weight	Weight	None
BMI	Body mass index	None
BMI4Class	BMI classification (4 categories)	-1 = '< 18.5', 0 = '18.5-24.9', 1 = '25-29.9', 2 = '≥30'
overweight	Overweight	0 = no, 1 = yes
obesity	Obesity	0 = no, 1 = yes
WGOCobBMI	WGOC BMI-based obesity	0 = Normal, 1 = Overweight, 2 = Obese
BMI25	CDS obesity (BMI≥25)	0 = 'BMI<25', 1 = 'BMI≥25'
Waist1	Classic waist circumference	None
WC9085	Abdominal obesity (JCD criteria)	0 = Normal, 1 = Obese
WC8580	WGOC waist circumference-based obesity	None
WC9080	IDF/NCEP-2005 obesity	0 = Normal, 1 = Obese
Waist2	Umbilical waist circumference	None
hip	Hip circumference	None
WHR	Waist-to-hip ratio	None
leg	Thigh circumference	None
HR	Heart rate	None
Bpsys	Systolic blood pressure	None
Bpdia	Diastolic blood pressure	None
BP13085	Abnormal blood pressure	0 = Normal, 1 = '≥13085'
WHO1999hbp	Hypertension	0 = Normal, 1 = Hypertension
Fat	Body fat percentage	None
Fat2Class	Body fat classification (2 categories)	0 = '< 25', 1 = '≥25'
Fat3Class	Body fat obesity	0 = Lean, 1 = Normal, 2 = Obese
GA	Glycated albumin (%)	None
FPG	Fasting plasma glucose	None
PG2h	2-hour postprandial glucose	None
HbA1c	Hemoglobin A1c (portable device)	None
HPLC	Hemoglobin A1c (HPLC method)	None
FINS	Fasting Insulin	None
FCP	Fasting C-peptide	None

INS2h	2-Hour Insulin	None
CP2h	2-Hour C-peptide	None
HomaIR	HOMA-IR	None
Homaβ	HOMA-β	None
ISIGutt	ISI Gutt	None
ALT	Alanine Aminotransferase	None
ALTn	ALT Segmentation	0 = '<40', 1 = '≥40'
AST	Aspartate Aminotransferase	None
ASTALT	AST/ALT	None
GGT	Gamma-Glutamyl Transferase (γGT)	None
ALP	Alkaline Phosphatase	None
TBILn	Total Bilirubin Elevation Segmentation	0 = '≤17.1', 1 = '> 17.1'
TBIL	Total Bilirubin	None
DBIL	Direct Bilirubin	None
IDBIL	Indirect Bilirubin	None
TP	Total Protein	None
ALB	Albumin	None
GLO	Globulin	None
AG	A/G	None
HbsAg	Hepatitis B surface antigen	0 = Negative, 1 = Positive
HCVAb	Hepatitis C antibody	0 = Negative, 1 = Positive
GLU	Fasting plasma glucose (biochemical analyzer)	None
CHOL	Total cholesterol	None
TG	Triglycerides	None
TGdis	Hypertriglyceridemia	0 = Normal, 1 = Abnormal
HDL	High-density lipoprotein	None
DysHDLIDFNCEP	Low HDL (IDF and NCEP-2005 criteria)	None
JCDHDLdis	HDL abnormality (cutoff 1.04 mmol/L)	0 = '≥1.04 mmol/L', 1 = '<1.04 mmol/L'
CDSdyelip	Dyslipidemia (CDS criteria)	0 = Normal, 1 = Abnormal
LDL	Low-density lipoprotein	None
BUN	Blood Urea Nitrogen (BUN)	None
SCRE	Serum Creatinine	None
HighCr	Renal Insufficiency	0 = 'None', 1 = 'Serum Creatinine > 106'
SUA	Serum Uric Acid	None
HighUA	Serum Uric Acid Elevation	0 = Normal, 1 = Elevated
UCRE	Urine Creatinine	None
UmALB	Urine Microalbumin	None
HighUmALB	Urine ALB Elevation	0 = Normal, 1 = Elevated
UMAUCR	Urine MA/UCR (Urine Microalbumin-to-Creatinine Ratio)	None
ACR3degree	ACR (Albumin-to-Creatinine Ratio) Three-Tier Classification	1 = '<30', 2 = '≥30 & <300', 3 = '≥300'
HighACR	ACR Elevation	0 = '<30', 1 = '≥30'
WBC	White Blood Cells (WBC)	None
LC	Percentage of Immature Cells	None
LCcount	Absolute Count of Immature Cells	None

ALY	Percentage of Atypical Lymphocytes	None
ALYcount	Absolute Count of Atypical Lymphocytes	None
RBC	Red Blood Cells (RBC)	None
RDWCV	RDW-CV	None
RDWSD	RDW-SD	None
MCV	Mean Corpuscular Volume (MCV)	None
HCT	Hematocrit (Hct)	None
LYMPH	LYMPH%	None
LYMPHcount	Lymphocyte Count	None
MCH	Mean Corpuscular Hemoglobin (MCH)	None
MCHC	Mean Corpuscular Hemoglobin Concentration (MCHC)	None
MPV	Platelet Mean Volume (MPV)	None
BASO	BASO%	None
BASOcount	Basophil Count	None
EO	EO%	None
EOcount	Eosinophil Count	None
HGB	Hemoglobin (Hb)	None
Hb4per	Hemoglobin 4 Classification	1 = '<90', 2 = '90-120', 3 = '120-150', 4 = '≥150'
PLT	Platelet (PLT)	None
PDW	PDW	None
PCT	PCT	None
MONO	MONO%	None
MONOcount	Monocyte Count	None
NEUT	NEUT%	None
NEUTcount	Neutrophil Count	None
ALT	None	None
FIB4	FIB-4 index	None
FIB4n	FIB-4 categorization	0 = '<1.45', 1 = '≥1.45'
PLCR	P-LCR	None
DMhis	History of Diabetes	0 = no, 1 = yes
CDSMS	CDS Metabolic Syndrome	0 = no, 1 = yes
DM	Diabetes	0 = no, 1 = yes
DMnew3class	Characteristics of newly diagnosed diabetes	1 = Diabetes diagnosed via fasting blood glucose, 2 = Diabetes diagnosed via 2-hour blood glucose, 3 = Diabetes diagnosed via both fasting and 2-hour blood glucose
PGclass3	Three-category classification of blood glucose	0 = NGR (Normal Glucose Regulation), 1 = 'Prediabetes, 2 = Diabetes Mellitus (DM)
PGclass6	Six-category classification of blood glucose	0 = NGR (Normal Glucose Regulation), 1 = IFG (Impaired Fasting Glucose), 2 = IGT (Impaired Glucose Tolerance), 3 = IFG & IGT (Combined Impaired

		Fasting Glucose and Impaired Glucose Tolerance), 4 = NDM (Neonatal Diabetes Mellitus), 5 = KDM (Ketosis-prone Diabetes Mellitus)
Anemia	Anemia	0 = normal, 1 = anemia
ThalussemiaAND	Thalassemia Initial Screening and	0 = normal, 1 = suspected
ThalussemiaOR	Thalassemia initial screening or	0 = normal, 1 = suspected
ThalussemiaHPLC	HPLC for thalassemia screening	1 = suspected
HypertenHis	History of Hypertension	0 = no, 1 = yes
DyslipideHis	History of Dyslipidemia	0 = no, 1 = yes
Smoking2	Categorized as 'no' for occasional smoking	0 = no, 1 = yes
Smoking	Smoking	1 = no, 2 = occasionally, 3 = regularly
SmokingAge	Age started smoking	None
SmokingRange	Years of smoking	None
Drinking2	Categorized as 'no' for occasional drinking	0 = no, 1 = yes
Drinking	Alcohol consumption	1 = none, 2 = occasionally, 3 = regularly
DrinkingAge	Age started drinking	None
Drink50g	Amount per occasion (50 g)	None
EscDrink	Ever quit drinking	None
EscDrinkM	How many months since quitting drinking	None
EscDrinkT	Number of times quit	None
EscDrinkR	Reason for quitting drinking	None
EscDrinkD	Name of the disease leading to quitting drinking	None
EscDrinkO	Other reasons for quitting drinking	None
DrinkYears	Total years of drinking	None
Tea	Tea consumption	1 = no, 2 = occasionally, 3 = regularly
Angina	History of Angina	0 = no, 1 = yes, 9 = unknown
Arrhythmia	History of Arrhythmia	0 = no, 1 = yes, 9 = unknown
PAD	History of PAD	0 = no, 1 = yes, 9 = unknown
Hyper16	History of Hyperthyroidism	0 = no, 1 = yes, 9 = unknown
Hyper13	Hyperuricemia	0 = no, 1 = yes, 9 = unknown
Blind	Blindness	0 = no, 1 = yes, 9 = unknown
Cancer	Cancer	0 = no, 1 = yes, 9 = unknown
CerebralHe	Intracerebral Hemorrhage	0 = no, 1 = yes, 9 = unknown

CerebralIn	Cerebral Infarction	0 = no, 1 = yes, 9 = unknown
TIA	History of TIA	0 = no, 1 = yes, 9 = unknown
MI	History of Myocardial Infarction	0 = no, 1 = yes, 9 = unknown
HF	History of Heart Failure	0 = no, 1 = yes, 9 = unknown
CAP	Carotid Artery Disease	0 = no, 1 = yes, 9 = unknown
Hypo14	Hypothyroidism	0 = no, 1 = yes, 9 = unknown
RF	Renal Failure	0 = no, 1 = yes, 9 = unknown
Amputation	Amputation	0 = no, 1 = yes, 9 = unknown
WeightLoss	Weight Loss of More Than 5 Jin in the Past Year	1 = no, 2 = yes, 9 = unknown
FatherDM	Paternal Diabetes	0 = no, 1 = yes, 9 = unknown
MotherDM	Maternal Diabetes	0 = no, 1 = yes, 9 = unknown
BSDM	Sibling Diabetes	0 = no
ChildDM	Offspring Diabetes	0 = no
DMfamilyHistory	Family History of Diabetes	0 = no, 1 = yes
Known	Awareness of Diabetic Symptoms	1 = no, 2 = yes, 9 = unknown
OPD	Obesity Predisposes to Diabetes	1 = no, 2 = yes, 9 = unknown
ToChilren	Genetic Risk of Diabetes in Offspring	1 = no, 2 = yes, 9 = unknown
DietExcise	Diet and Exercise for Diabetes Control	1 = no, 2 = yes, 9 = unknown
Compl	Diabetic Complications	1 = no, 2 = yes, 9 = unknown
Check	Blood Glucose Testing History	1 = no, 2 = yes, 9 = unknown
GlucoseLev	Awareness of Blood Glucose	1 = no, 2 = yes, 9 = unknown
DF	History of Diabetic Foot	2 = yes
MADPN	Monofilament Assessment of DPN	0 = no, 1 = yes
TFDPN	Diabetic peripheral neuropathy (DPN) assessed by tuning fork test	0 = no, 1 = yes
DR	Diabetic Retinopathy	1 = yes
EyeDia	Ophthalmology Diagnosis Summary	None
GAnum	Population size of the Jinuo ethnic group undergoing Glycated Albumin (GA) testing	None
filter	age >= 18 & age <= 75 & UMAUCR >= 0 & GFR5lev >= 0 & Fasting Plasma Glucose (FPG) >= 0 & HbA1c >= 0 & smoking >= 0 & drinking >= 0 & Bpsys	0 = Not Selected, 1 = Selected

	≥ 0 & BMI > 1 & CDSdyelip ≥ 0 (FILTER)	
GAstudy	Diagnosis Based on Meeting All Three Criteria (No DM History)	None
A1cover6575	HbA1c is divided into three categories	1 = '< 6.5', 2 = '6.5 to 7.5', and 3 = '> 7.5'
DMhighrisk	High-risk population for diabetes	1 = yes
FPGover7	Fasting plasma glucose ≥ 7.0 mmol/L	0 = no, 1 = yes
P2PGover11	2-hour postprandial glucose ≥ 11.1 mmol/L	0 = no, 1 = yes
DRyd	Fundus Photography with Possible DR	1 = yes
cataract	Cataract	1 = yes
GFR	Calculated GFR (Glomerular Filtration Rate)	None
GFR5lev	GFR Five-Tier Classification (GFR5lev)	1 = ' ≥ 90 ', 2 = '<90', 3 = '<60', 4 = '<30', 5 = '<15'
GFR6levG5	GFR3a3b	1 = 'eGFR ≥ 90 mL/min/1.73m ² ', 2 = 'eGFR <90 mL/min/1.73m ² & ≥ 60 mL/min/1.73m ² ', 4 = 'eGFR <30 mL/min/1.73m ² & ≥ 15 mL/min/1.73m ² ', 5 = 'eGFR <15 mL/min/1.73m ² ', 31 = 'eGFR <60 mL/min/1.73m ² & ≥ 45 mL/min/1.73m ² ', 32 = 'eGFR <45 mL/min/1.73m ² & ≥ 30 mL/min/1.73m ² '
EUGFRabACR012	eGFR normal urine ACR segmentation	0 = 'eGFR ≥ 90 mL/min/1.73m ² and ACR < 30', 1 = 'eGFR ≥ 90 mL/min/1.73m ² and ACR ≥ 30 ', 2 = 'eGFR < 90 mL/min/1.73m ² '
EUGFR90UACR01	eGFR less than 60 mL/min/1.73m ² or UACR greater than or equal to 30mg/g	0 = 'eGFR ≥ 60 mL/min/1.73m ² and UACR < 30 mg/g', 1 = 'eGFR < 60 mL/min/1.73m ² or UACR ≥ 30 mg/g'
EUGFR60abACR012	eGFR ≥ 60 normal urine ACR segmentation	0 = 'eGFR ≥ 60 mL/min/1.73m ² and ACR < 30', 1 = 'eGFR ≥ 60 mL/min/1.73m ² and ACR ≥ 30 ', 2 = 'eGFR < 60 mL/min/1.73m ² '
UACRGFR	Positivity defined as urine ACR ≥ 30 and eGFR < 60 mL/min/1.73m ²	0 = 'Negative', 1 = 'Positive'
DN	Diabetic nephropathy staging	0 = 'Urine < 30 mg', 3 = 'Stage 3 ≥ 30 mg', 4 =

		'Stage 4 ≥ 300 mg with GFR < 10 mL/min/1.73m ² ', 5 = 'Stage 5 uremia'
--	--	---